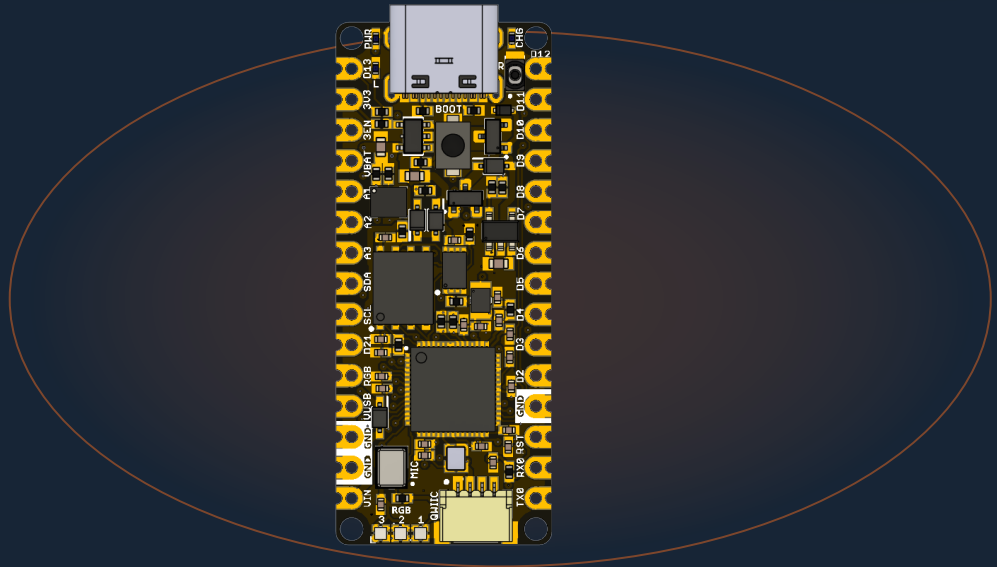


MEMBER OF THE UNIT DEVLAB ECOSYSTEM

UNIT PULSAR RP2350 SENSE

MPN: UE0103

Version 0.1.0



RP2350A development board with external flash and PSRAM, motion sensing, PDM audio, microSD, QWIIIC, USB-C, and HSTX expansion.

RP2350A

16 MiB FLASH

8 MiB PSRAM

SDIO · PDM · I2C · HSTX

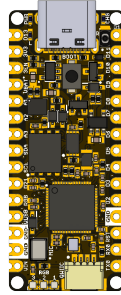
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Description

The UNIT PULSAR RP2350 is a multi-interface development board in the UNIT DevLab ecosystem, built around the Raspberry Pi RP2350A microcontroller. The V1.3 design combines external QSPI flash and PSRAM with motion sensing, PDM audio, microSD, USB-C, battery support, addressable RGB indicators, QWIIC I2C, and a 22-pin HSTX expansion connector.



Applications

- RP2350 firmware and peripheral prototyping
- Motion and PDM audio acquisition
- microSD data logging
- HSTX graphics experiments
- External-memory and multicore applications
- Education and subsystem validation

Hardware Features

- RP2350A microcontroller with external W25Q128 128 Mbit QSPI flash
- APS6404L-3SQR-ZR 8 MiB PSRAM
- BMI270 six-axis inertial measurement unit
- ICS-41350 digital PDM microphone
- Four-bit SDIO microSD connection
- Three WS2812-compatible RGB LEDs plus power, charge, and user indicators
- USB-C, QWIIC, battery, edge-pad, SWD, and HSTX connections

The Board

The UNIT PULSAR RP2350 is a development board for evaluating the Raspberry Pi RP2350A and building applications that combine high-speed digital output, removable storage, motion sensing, digital audio, external memory, and general-purpose I/O. Hardware revision V1.3.0 places the controller, QSPI flash, PSRAM, power components, indicators, and user controls on the top side. The microSD socket, battery connection, onboard microphone, debug pads, and HSTX connector are accessible from the bottom side.

1.1 Accessories

The following optional accessories are recommended for use with the UNIT PULSAR RP2350A. Select accessories according to the interfaces and features required by the application.

Accessory	Purpose	Selection notes
QWIIC cable	External I2C expansion	4-pin, 1 mm pitch cable for connecting compatible 3.3 V QWIIC devices
USB-C data cable	Power, programming, and USB serial	Must support data; a charge-only cable cannot upload firmware
MicroSD card	Removable storage and data logging	Class 10 microSD card; FAT32 formatting is recommended for the provided examples
UNIT DevLab DVI to FPC Adapter	HSTX DVI-compatible video output	Recommended adapter for using the board's HSTX video interface through the 22-pin FPC connector
22-pin FPC cable	Connection between the HSTX connector and DVI adapter	22-position, 0.5 mm pitch; planned for future availability from UNIT Electronics
LiPo 3.7 V 650 mAh Battery	Battery-powered operation	Recommended single-cell LiPo battery; verify polarity and connector orientation before connection

Verify connector type, orientation, polarity, voltage domain, and pinout before connecting accessories.

1.2 Board Identification

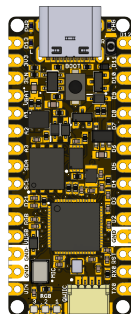
Item	Value
Product	UNIT PULSAR RP2350
Product family	UNIT DevLab ecosystem
Product type	Multi-interface RP2350A development board
Main component	Raspberry Pi RP2350A, QFN-60
Hardware revision	V1.3.0
Product Reference	Version 0.1.0
Primary programming interface	USB-C / RP2350 USB boot workflow
Debug interface	SWD pads on bottom side

Hardware revision and documentation revision are identified independently.

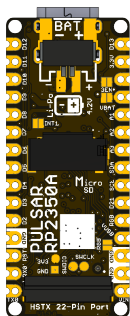
1.3 Main Assemblies

RefDes	Component	Function
IC3	RP2350A	Main processor and peripheral controller
IC1	W25Q128JVP1Q	128 Mbit (16 MiB) external QSPI flash
IC4	APS6404L-3SQR-ZR	8 MiB external PSRAM
IC5	BMI270	Six-axis accelerometer and gyroscope
MK1	ICS-41350	Digital PDM microphone
U1	AP2112K-3.3TRG1	Fixed 3.3 V LDO
IC2	MCP73831T-2ACI/OT	Single-cell Li-Ion/Li-Polymer charge controller
XTAL1	XOS20012000LT00351005	12 MHz reference oscillator
MICRO_SD-HOLDER	47309-2651	Removable microSD storage
LED1-LED3	WS2812 1010	Cascaded addressable RGB indicators
J1	HCZZ0032-4	Four-position QWIIC-style I2C connector
J5	FH34SRJ-22S-0.5SH(50)	22-position HSTX FFC/FPC connector
JP1	PH2.0 2P	Two-position battery connection

1.4 Board Views



The top view identifies the USB-C connector, BOOT and reset controls, edge-pad labels, RP2350A, flash, PSRAM, QWIIC connector, oscillator, and three RGB LEDs.



The bottom view identifies the battery polarity marks, microSD socket, microphone acoustic port, SWD pads, HSTX connector, and V1.3.0 revision marking.

1.5 Handling

Handle the board using normal ESD precautions. Avoid touching the microphone port, connector contacts, or exposed test pads. Remove power before inserting or removing FFC/FPC and battery connectors. Keep conductive objects away from the bottom-side battery and microSD areas when the board is energized.

2 Ratings

This chapter defines the recommended operating conditions and electrical characteristics of the UNIT PULSAR RP2350A hardware V1.3.0. Ratings apply to the complete board unless otherwise stated.

2.1 Recommended Operating Conditions

The UNIT PULSAR RP2350A can be powered from USB-C, VIN, or a single-cell LiPo battery. The onboard AP2112K regulator supplies the 3.3 V logic domain used by the RP2350A and onboard peripherals.

Parameter	Recommended operating condition	Notes
USB-C input	5 V nominal	Primary power and programming interface
VIN input	Up to 6 V	External board supply input
Battery input	3.7 V nominal, 4.2 V maximum	Single-cell LiPo battery
Regulated logic rail	3.3 V nominal	Generated by the onboard AP2112K-3.3 regulator
3.3 V regulator output capability	Up to 600 mA	Shared by the RP2350A, onboard peripherals, LEDs, and external 3.3 V loads
Battery charge current	200 mA nominal	Configured by the MCP73831 PROG resistor
RP2350 system clock	Up to 150 MHz	Firmware configurable
Board reference oscillator	12 MHz	Onboard reference oscillator
Internal SRAM	520 kB	RP2350A internal memory
External flash	128 Mbit / 16 MiB	W25Q128JVP1Q
External PSRAM	8 MiB	AP56404L-3SQR-ZR
Logic level	3.3 V	Applies to onboard digital interfaces and GPIO

2.2 Power Domains and Power Distribution

The UNIT PULSAR RP2350A uses separate power domains for external supply, battery operation, system power, and regulated 3.3 V logic.

The main power domains are:

- **USB-C / VBUS:** provides the nominal 5 V USB supply used for board power, programming, and battery charging.
- **VIN:** external supply input supporting up to 6 V.
- **VBAT:** battery domain intended for a single-cell LiPo battery with a nominal voltage of 3.7 V and a maximum fully charged voltage of 4.2 V. The battery input includes reverse-polarity protection.
- **VSYS:** internal system supply used as the input to the 3.3 V regulator. The board power-selection circuitry routes the available supply source to this domain.

- 3.3 V:** regulated logic supply generated by the AP2112K-3.3. This rail powers the RP2350A, external flash, PSRAM, BMI270, PDM microphone, microSD interface, QWIIC interface, RGB LEDs, and other onboard 3.3 V circuitry.

The AP2112K-3.3 regulator supports up to 600 mA output current. This current is shared between the RP2350A, onboard peripherals, status and RGB LEDs, and any external device powered from the 3.3 V rail. The current available to external loads therefore depends on the active board configuration and application workload.

The MCP73831T-2ACI/OT manages single-cell LiPo battery charging. Hardware V1.3.0 is configured for a nominal charge current of 200 mA. The charging current is determined by the PROG resistor and may be changed by modifying its value according to the MCP73831 configuration requirements.

All onboard digital interfaces operate in the 3.3 V logic domain. QWIIC, microSD, HSTX, and exposed GPIO signals must therefore be used with 3.3 V-compatible devices unless external level translation is provided.

Connector mechanical current ratings must not be interpreted as board power ratings. Maximum usable current is also constrained by the associated PCB traces, power path, regulator, protection components, and operating conditions.

2.3 Digital Interfaces

All onboard digital peripherals are connected to the board's 3.3 V logic domain. External logic compatibility must nevertheless be checked against the RP2350A input/output specifications and the power state of both devices.

Interface	Board connection	Electrical note
Internal I2C	GPIO8 / GPIO9	BMI270 bus with onboard pull-ups
QWIIC I2C	GPIO24 / GPIO25	Connector also supplies 3.3 V and GND
microSD	GPIO2–GPIO7	Full SDIO group; wiki uses SPI-compatible subset
PDM	GPIO10 / GPIO11	Clock output and microphone data input
HSTX	GPIO12–GPIO19	High-speed driven pairs; reserve while video is active
WS2812 data	GPIO1	Drives three cascaded onboard LEDs
SWD	SWDIO / SWCLK	Use a 3.3 V target reference

2.4 Confirmed Component Values

Symbol / parameter	Value	Scope
RP2350 internal SRAM	520 kB	RP2350A component
RP2350 system clock	150 MHz maximum nominal	RP2350A component
QSPI flash capacity	128 Mbit	IC1
PSRAM capacity	8 MiB	IC4
PDM microphone supply	1.65 to 3.63 V	ICS-41350 component only
LDO output	3.3 V nominal	U1 component selection
Oscillator frequency	12 MHz	XTAL1

2.5 Application-Dependent and Uncharacterized Parameters

Some electrical and performance characteristics depend on firmware configuration, operating mode, connected peripherals, and application load. Unless otherwise specified, these parameters should not be interpreted as guaranteed board-level limits.

- **Available 3.3 V current for external loads:** the AP2112K-3.3 provides up to 600 mA total output current. This current is shared by the RP2350A, onboard memories, sensors, LEDs, and external 3.3 V loads. Available external current therefore depends on the active board configuration and application workload.
- **Board current consumption:** total current consumption varies with RP2350A activity, clock configuration, enabled peripherals, RGB LED brightness, external memory activity, microSD operation, audio capture, HSTX video output, and connected external devices.
- **High-speed interface performance:** maximum validated operating rates for external microSD, PDM, I2C, and HSTX interfaces depend on firmware configuration, connected devices, cabling, and signal integrity.
- **GPIO electrical characteristics:** GPIO drive strength, input thresholds, and other pin-level electrical characteristics follow the RP2350A specifications unless a board-level circuit modifies the corresponding signal.
- **Environmental characteristics:** board-level operating temperature, humidity, ESD immunity, and thermal performance are not specified unless explicitly stated in a released qualification or validation specification.

2.6 Electrical Precautions

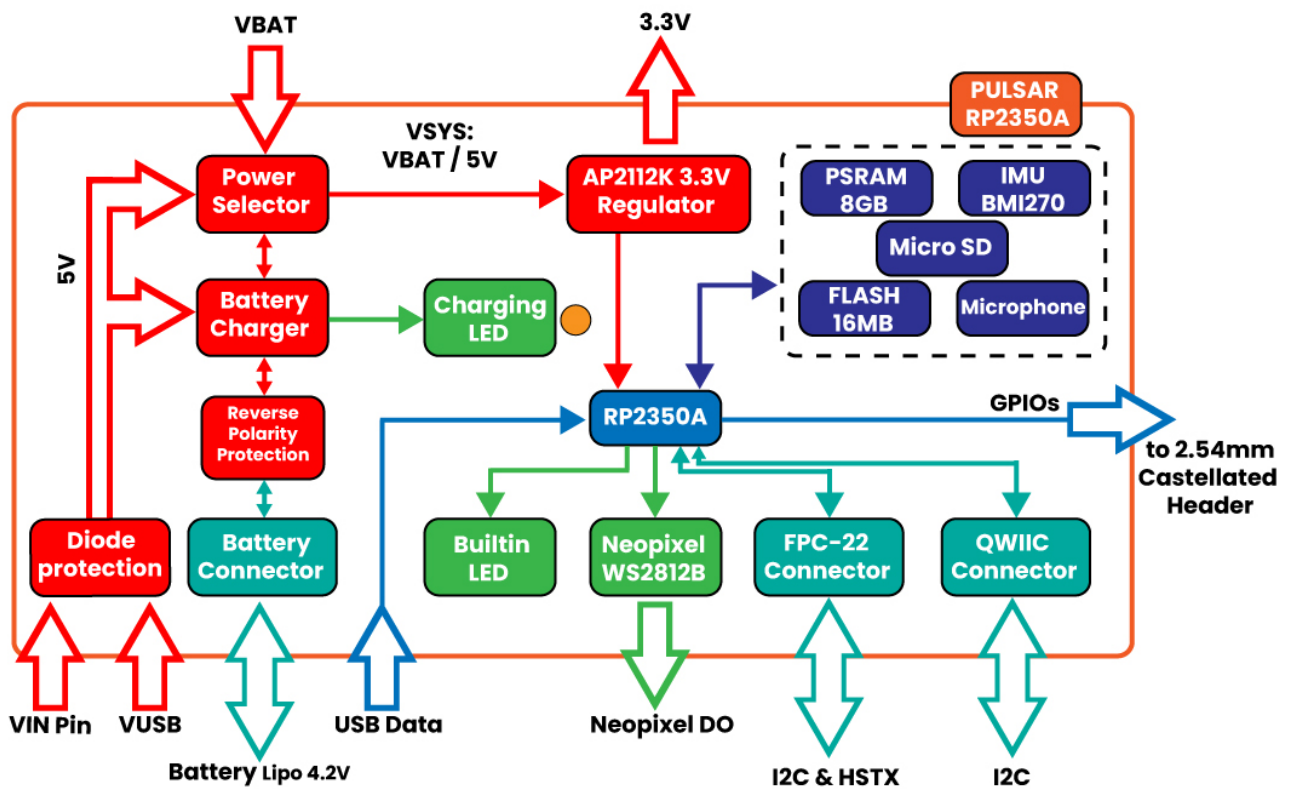
1. Use only the specified power-input ranges: 5 V nominal through USB-C, up to 6 V through VIN, or a 3.7 V nominal / 4.2 V maximum single-cell LiPo battery.
2. All GPIO and onboard digital interfaces operate in the 3.3 V logic domain. **Do not apply 5 V directly to GPIO, QWIIIC, microSD, HSTX, SWD, or other 3.3 V signal lines.**
3. Observe the battery connector polarity before connecting a single-cell LiPo battery. Do not connect multi-cell battery packs or batteries exceeding 4.2 V.
4. Do not externally drive the 3.3 V rail while the onboard regulator is active unless the external power source and operating configuration have been specifically designed for this purpose.
5. Connect external circuits to a common ground before applying external clock, data, or control signals.
6. Do not connect externally driven circuits to GPIO12–GPIO19 while the HSTX interface is active, as these GPIOs are used by the HSTX video output.
7. Remove power before inserting, removing, or changing FFC/FPC or battery connections. Avoid inserting or removing the microSD card while a write operation is in progress.
8. Ensure that the total load on the 3.3 V rail, including onboard circuitry and externally connected devices, remains within the 600 mA capability of the AP2112K-3.3 regulator.

3 Functional Overview

The UNIT PULSAR RP2350 is organized around the RP2350A processor, with dedicated external memories and fixed onboard peripherals. Firmware can use each block independently or combine acquisition, storage, user feedback, and video in one application.

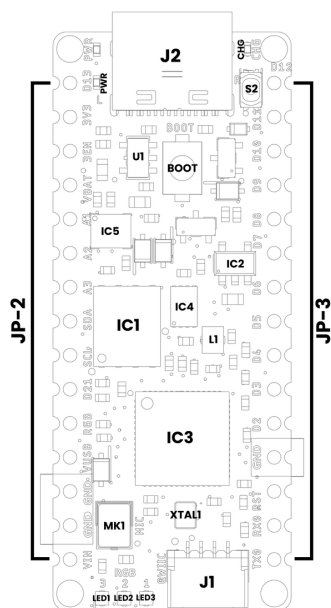
3.1 Block Diagram

PULSAR RP2350A Block Diagram

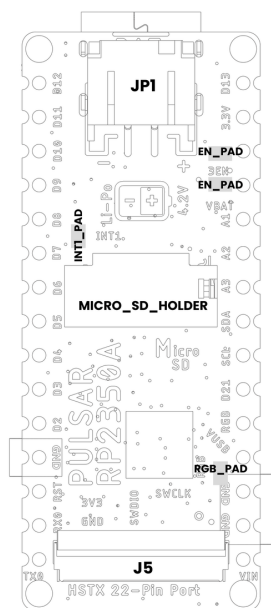


The RP2350A controls every data path. W25Q128 flash stores executable code and nonvolatile application data. PSRAM extends working memory. The microSD socket provides removable storage. BMI270 and ICS-41350 devices provide motion and audio input. QWIIC and edge pads expose expansion signals, while HSTX and the LEDs provide high-speed and visual outputs.

3.2 Board Topology



Top Side



Bottom Side

Views of Board Topology

RefDes	Component	Confirmed role
IC3	RP2350A	Main microcontroller
IC1	W25Q128JVP1Q	128 Mbit (16 MiB) QSPI flash memory
IC4	APS6404L-3SQR-ZR	8 MiB external PSRAM
IC5	BMI270	Six-axis IMU connected to the internal I2C bus
MK1	ICS-41350	Digital PDM MEMS microphone
U1	AP2112K-3.3TRG1	Fixed 3.3 V LDO regulator
IC2	MCP73831T-2ACI/OT	Single-cell Li-Ion/LiPo battery charge controller
XTAL1	12 MHz oscillator	RP2350A reference clock source
L1	3.3 μ H inductor	RP2350A internal switching-regulator inductor
MICRO_SD_HOLDER	47309-2651	microSD card socket connected through four-bit SDIO signals
LED1-LED3	XL-1010RGBC-WS2812B	Cascaded addressable RGB LEDs

RefDes	Component	Confirmed role
J1	HCZZ0032-4	Four-position, 1 mm pitch QWIIC connector
J2	USB-C connector	USB power, programming, and USB data interface
JP1	Battery connector	Two-position single-cell LiPo battery connection
J5	FH34SRJ-22S-0.5SH(50)	22-position, 0.5 mm pitch FFC/FPC connector for HSTX video and auxiliary signals
S2	BOOT switch	Forces RP2350A USB boot mode during startup/reset
JP2	Castellated header row	Left-side GPIO, power, analog, and digital expansion
JP3	Castellated header row	Right-side GPIO, power, analog, and digital expansion
EN_PAD	Solder jumper	Connects the AP2112K enable signal to the castellated header, allowing external control of the 3.3 V regulator
VBAT_PAD	Solder jumper	Connects the VBAT rail to the castellated header, providing access to the battery supply
INT_PAD	Solder jumper	Connects the BMI270 interrupt signal to the RP2350A GPIO, enabling hardware interrupt operation
RGB_PAD	Solder jumper	Connects the data output of the third onboard NeoPixel to the castellated header, allowing the RGB chain to be extended with external NeoPixels
SWDIO	Debug test pad	RP2350A Serial Wire Debug data signal
SWCLK	Debug test pad	RP2350A Serial Wire Debug clock signal
GND	Debug test pad	Ground reference for the SWD interface

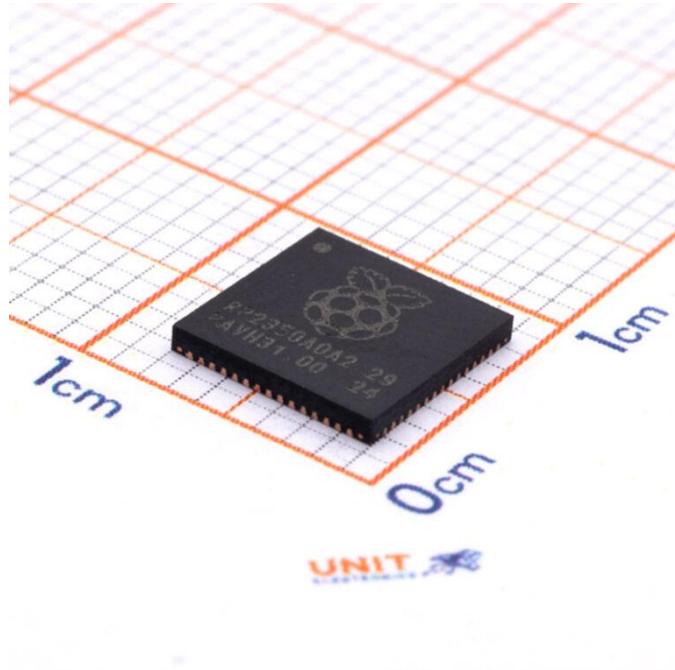
The top side groups programming, processing, memory, and user-visible controls. USB-C, BOOT, reset, QSPI flash, PSRAM, RP2350A, QWIIC, and RGB indicators are visible from this side. The bottom side groups removable storage, battery, audio, SWD, and HSTX connections.

Board region	Main elements	Design intent
USB end	USB-C, BOOT, reset, power/charge indicators	Programming and initial power-up
Upper center	power-path devices and PSRAM	Source routing and working-memory expansion
Center	RP2350A and 12 MHz oscillator	Processing and clock generation
Lower center	QSPI flash and PDM microphone area	boot storage and audio acquisition
QWIIC end	three RGB LEDs and QWIIC connector	status display and external I2C
Bottom side	battery, microSD, BMI270, SWD, HSTX	storage, sensing, debug, and video expansion

The two edge-pad rows expose power, analog, serial, and general-purpose signals. Some HSTX-routed signals are shared with edge positions, so firmware ownership must be decided before enabling high-speed video.

3.3 Processor

RP2350A is the main controller. According to the Raspberry Pi component datasheet, the device provides two selectable processor architectures: dual Arm Cortex-M33 or dual Hazard3 RISC-V cores, with a nominal system frequency up to 150 MHz. Firmware selects one architecture for a given build; it does not execute Arm and RISC-V code simultaneously.



Raspberry Pi RP2350A QFN-60 microcontroller

The processor integrates 520 kB of on-chip SRAM arranged in ten banks, USB 1.1 host/device control with an embedded PHY, ADC inputs shared with GPIO, fixed serial peripherals, PWM, DMA, three PIO blocks with twelve state machines, and one HSTX peripheral. The PULSAR design routes a selected subset of these functions to onboard devices and external connectors.

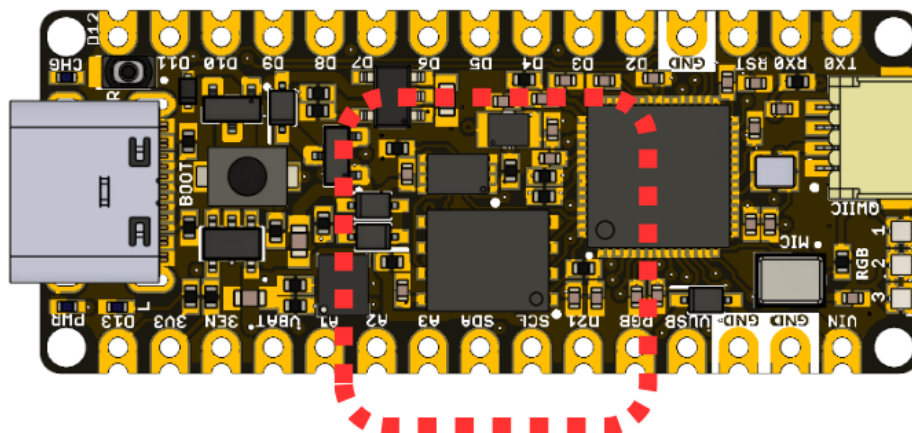
The on-chip SRAM is the lowest-latency working memory. It contains stacks, heaps, interrupt data, time-critical buffers, and code copied from flash when a framework requires it. DMA and multicore applications must coordinate access to shared buffers and peripherals.

3.4 Memory Architecture: Flash and PSRAM

The W25Q128JVPIQ provides 128 Mbit (16 MiB) of external QSPI flash. It stores the boot image, application firmware, and any filesystem or data region defined by the selected board-package partition scheme. The BOOT control participates in the RP2350 USB boot workflow used to load recovery or update images.

The APS6404L-3SQR-ZR provides 8 MiB of external PSRAM with chip select on GPIO0. In the Arduino workflow, the board configuration must set PSRAM CS to GPIO0 and select the 8 MiB capacity. Static objects can be placed in PSRAM with the core's PSRAM attribute; dynamic objects can be allocated with `pmalloc()` and released with `free()`.

PSRAM



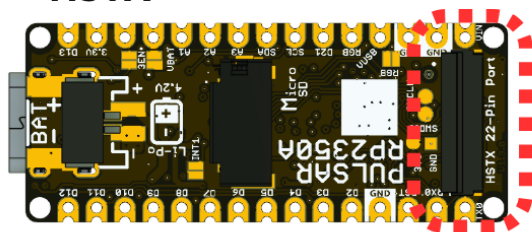
External PSRAM location

PSRAM is suited to framebuffers, file buffers, captured audio, sensor history, and large application structures. Interrupt flags, frequently accessed state, and timing-critical data should remain in internal SRAM. The application must check allocation results and avoid assuming that all configured PSRAM remains free after global/static allocations.

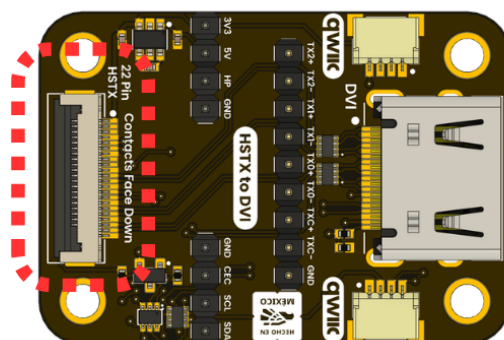
3.5 HSTX Video Output

The HSTX peripheral serializes data at high speed and can generate the pseudo-differential TMDS signals used by DVI-compatible displays. The wiki and PCB routing define four channel pairs:

HSTX



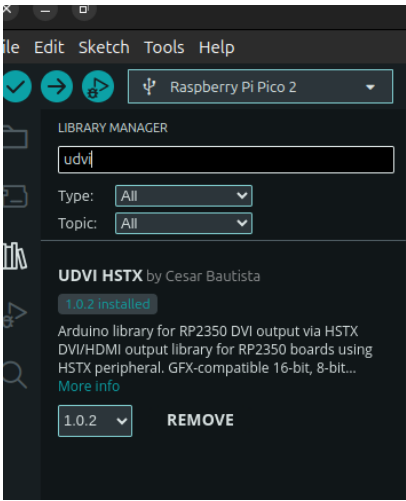
HDMI/DVI Video



HSTX connector location

TMDS channel	Positive GPIO	Negative GPIO	Function
Clock	14	15	Pixel/link clock pair
Data 0	18	19	TMDS data channel 0
Data 1	16	17	TMDS data channel 1
Data 2	12	13	TMDS data channel 2

The 22-position HSTX connector carries these signals and returns. The wiki uses the UDVI HSTX library with a 320 × 240 RGB565 framebuffer, drawing primitives, text, and partial-screen updates. This resolution is an implementation profile from the wiki, not a limit of the connector or RP2350A.



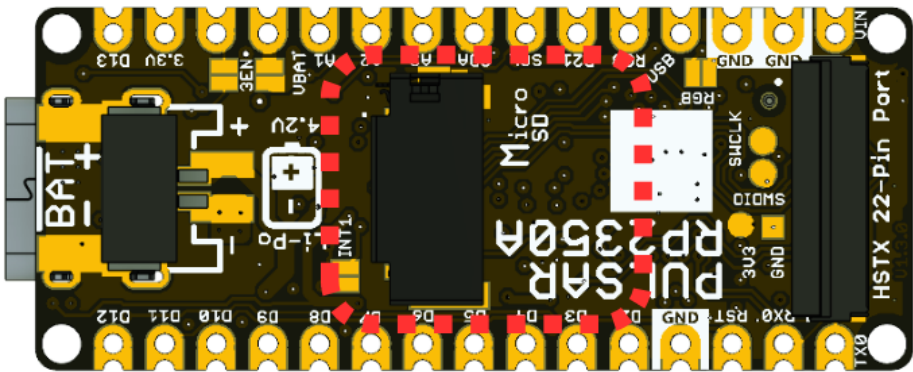
UDVI HSTX library in Arduino Library Manager

While HSTX is enabled, GPIO12–GPIO19 are actively driven and must not be shared with another output or an attached circuit that drives the same nets. Display adapters must match the connector contact order and differential-pair routing.

3.6 MicroSD Card Socket

The onboard 47309-2651 microSD socket is connected to a complete four-bit SDIO signal group and operates in the 3.3 V logic domain.

MicroSD



microSD socket location

Signal	GPIO	SPI-mode role in current examples
SDIO_CLK	2	SCK
SDIO_CMD	3	MOSI
SDIO_DAT0	4	MISO

Signal	GPIO	SPI-mode role in current examples
SDIO_DAT1	5	Not used by SPI mode
SDIO_DAT2	6	Not used by SPI mode
SDIO_DAT3	7	Chip select

The provided examples use the SPI-compatible subset through SPI and SDFS for initialization, file creation, reading, directory enumeration, and data logging.

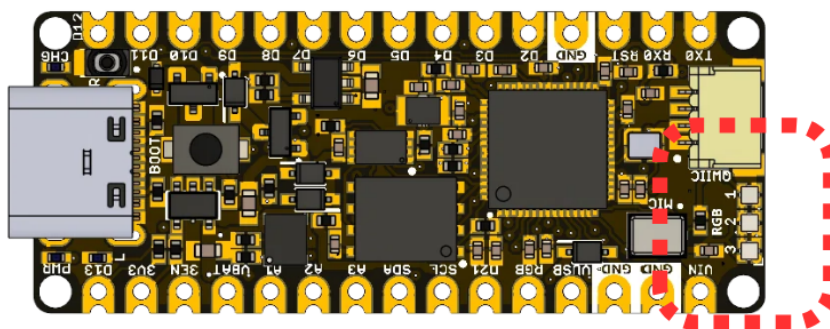
Software with four-bit SDIO support can additionally use DAT1 and DAT2. File writes should be flushed and closed before card extraction or power removal.

Card capacity, speed class, and filesystem support depend on the software environment. FAT32 is recommended for the provided examples.

3.7 LED Indicators

Three XL-1010RGBC-WS2812B addressable RGB LEDs form a serial chain driven by GPIO1. Firmware transmits an ordered color frame through the chain, allowing the three onboard pixels to provide status indication, progress animation, and user feedback from a single GPIO.

WS2812B RGB LED

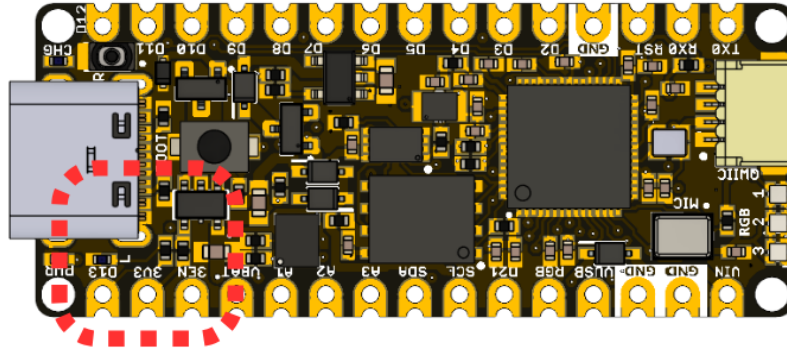


WS2812 RGB LED locations

The RGB_PAD solder jumper can connect the data output of the third onboard RGB LED to the castellated header. When this jumper is closed, the onboard RGB chain can be extended with compatible external addressable RGB LEDs.

The board also includes a user indicator on GPIO22 (D13 / BUILTIN1), a power indicator, and a battery charge-status indicator. The user LED and RGB LEDs are firmware-controlled. The power indicator follows the corresponding power rail, while the charge-status indicator is controlled by the MCP73831 STAT output.

LED BUILTIN



Built-in user LED location

Applications should limit RGB brightness when power consumption or thermal rise is important. RGB LED control is independent of the HSTX video interface.

3.8 AP2112K and MCP73831 Power Management System

U1 is an AP2112K fixed 3.3 V LDO regulator. It converts the VSYS rail into the regulated 3.3 V logic domain used by the RP2350A, external memories, sensors, indicators, microSD interface, and QWIIC connector. The regulator supports up to 600 mA total output current, shared between the onboard circuitry and any external load connected to the 3.3 V rail.

The AP2112K EN signal controls the 3.3 V regulator. The EN_PAD solder jumper allows this signal to be routed to the castellated header, providing external control of the regulator enable function when the jumper is closed.

IC2 is an MCP73831T-2ACI/OT single-cell Li-Ion/LiPo battery charge controller. Hardware V1.3.0 is configured for a nominal charge current of 200 mA. The charge current is established by the PROG resistor and may be modified by changing this resistor according to the MCP73831 configuration requirements.

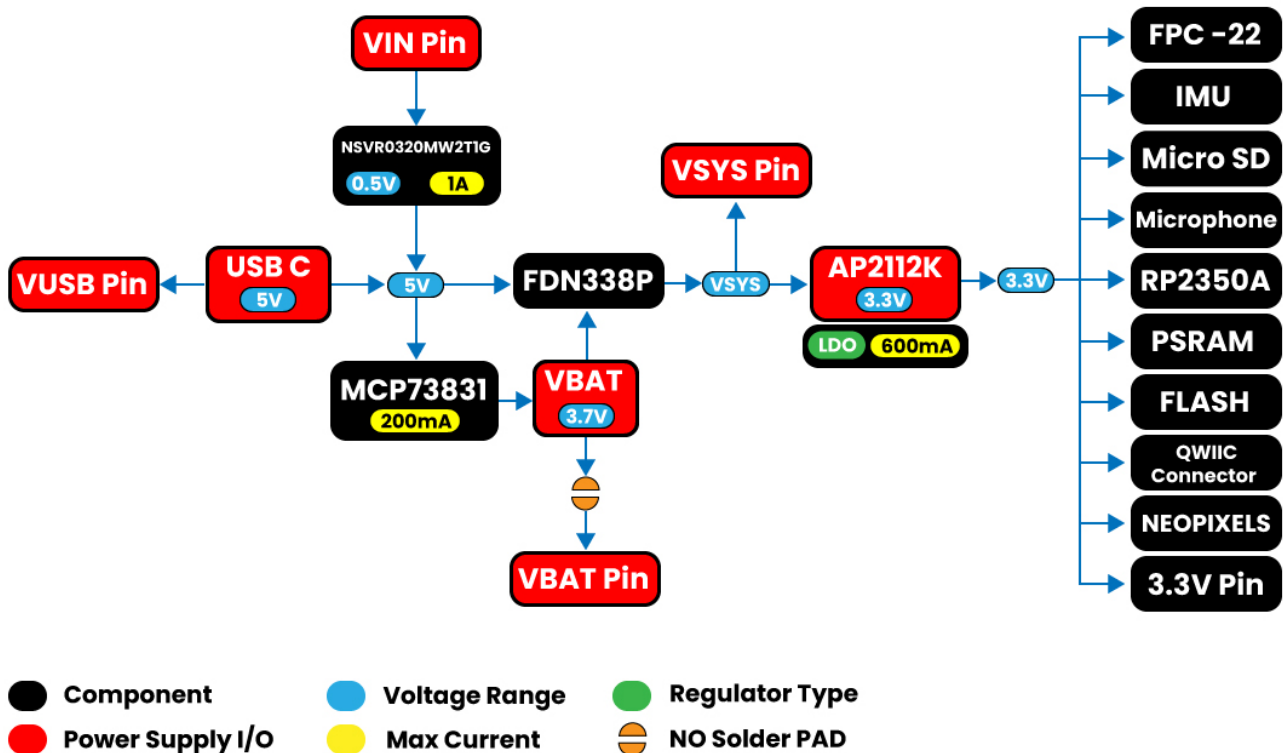
The battery input is intended for a single-cell LiPo battery with a nominal voltage of 3.7 V and a maximum fully charged voltage of 4.2 V. The battery path includes reverse-polarity protection.

The VBAT_PAD solder jumper allows the battery rail to be connected to the castellated header. When closed, the battery voltage can be accessed directly from the corresponding header position for external circuitry.

The board power-path circuitry combines USB, VIN, battery, VSYS, and the regulated 3.3 V domain. Schottky diodes and MOSFETs provide source routing and battery protection according to the implemented hardware design.

3.9 Power Tree

PULSAR RP2350A Power Tree



The power tree summarizes the main supply domains and their relationship within the UNIT PULSAR RP2350A. It is intended as a functional reference and does not replace the complete schematic.

The board can be powered from USB-C, VIN, or a single-cell LiPo battery. These sources are routed through the onboard power-path circuitry to the VSYS domain. The AP2112K-3.3 then generates the regulated 3.3 V rail used by the RP2350A and onboard peripherals.

The 3.3 V rail is also available through the castellated headers and QWIIC connector. The AP2112K provides up to 600 mA total output current. This current is shared between the RP2350A, memories, sensors, LEDs, microSD interface, and externally connected 3.3 V devices.

The current available to external loads therefore depends on the active onboard circuitry and application workload. The total 3.3 V load must remain within the regulator output capability.

The VBAT_PAD and EN_PAD solder jumpers provide optional access to the battery rail and regulator-enable control through the castellated headers.

3.10 BMI270 Motion Sensor

The onboard BMI270 combines a three-axis accelerometer and three-axis gyroscope. It is connected to GPIO8 (SDA) and GPIO9 (SCL) on the internal I2C bus. Firmware can detect the sensor through its supported I2C address configuration and read acceleration and angular-rate data.

The INT_PAD solder jumper provides an optional hardware interrupt connection between the BMI270 and GPIO12 of the RP2350A. When the jumper is closed, BMI270 interrupt events can be routed directly to GPIO12 for interrupt-driven applications without continuous polling.

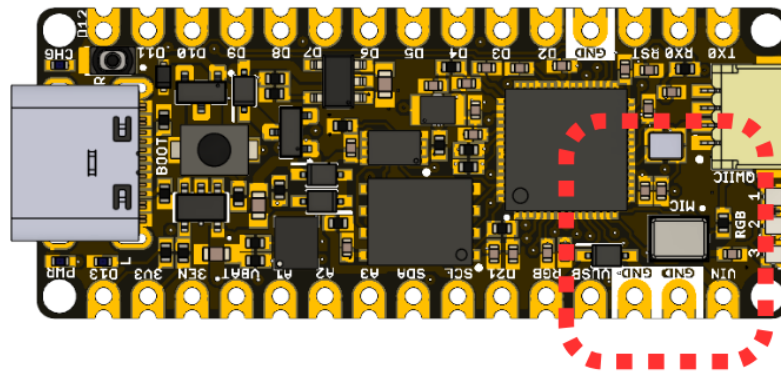
Typical applications include orientation interfaces, motion-triggered logging, gesture input, vibration monitoring, and motion-controlled graphics or user interfaces.

Measurement range, filtering, output data rate, interrupt source, and interrupt behavior are configured through the BMI270 registers or software driver and are not fixed by the PCB.

3.11 PDM Microphone

The ICS-41350 is an onboard digital MEMS microphone. RP2350A supplies its PDM clock on GPIO10 and receives the one-bit data stream on GPIO11. The software PDM stack clocks the microphone, decimates the stream, and produces PCM sample buffers for analysis, visualization, streaming, or storage.

PDM Microphone



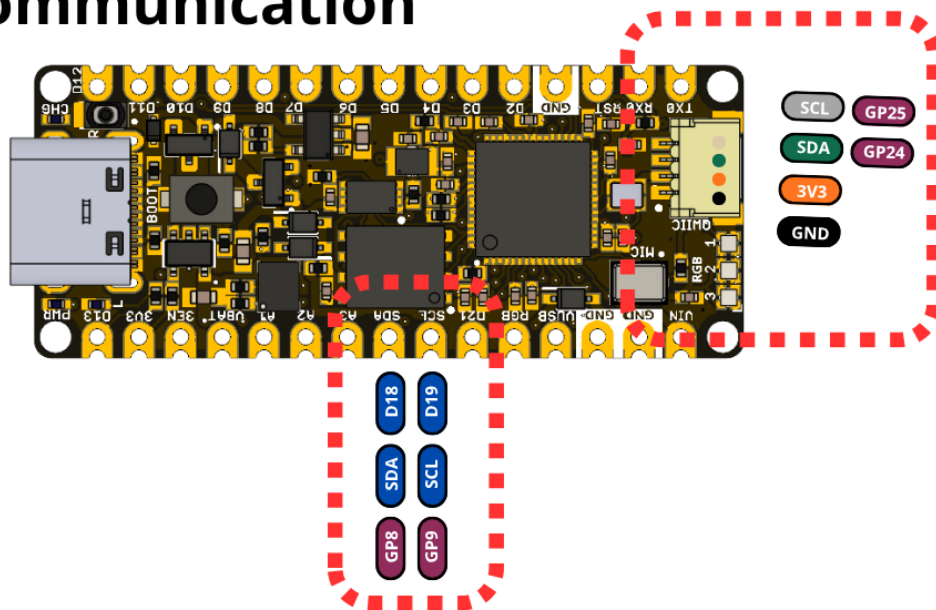
PDM microphone location

The acoustic port must remain unobstructed. Enclosures, adhesive, contamination, and board mounting can affect acoustic performance. Board-level sensitivity, noise, frequency response, and enclosure behavior depend on the mechanical and acoustic implementation and are not specified at module level.

3.12 I2C and QWIIC Expansion

The board provides two distinct I2C routes. GPIO8/GPIO9 serve the onboard BMI270 and are also represented by SDA/SCL edge labels. GPIO24/GPIO25 feed the four-position QWIIC connector and HSTX connector positions.

I²C Communication



I2C and QWIIC connections

This separation lets an application keep the IMU bus independent from external QWIIC sensors. The wiki includes bus scanning, BMI270 access, EEPROM byte and block operations, hexadecimal dumps, and structured EEPROM records. External pull-ups, device addresses, bus capacitance, and cable length must be considered when attaching multiple devices.

3.13 Combined System Operation

The board is designed for concurrent use of its subsystems. For example, the BMI270 can control an HSTX-rendered object while PSRAM holds large graphics or history buffers; the microphone or an I2C sensor can produce records stored on microSD; and RGB LEDs can show acquisition, storage, or fault state.

A robust application initializes one subsystem at a time, checks every return value, keeps high-rate and interrupt data in internal SRAM, moves large buffers to PSRAM, flushes storage before shutdown, and yields enough execution time for USB and framework services. The repository's C++ examples implement each stage separately before the combined application.

4 Connectors & Pinouts

Pin labels describe hardware revision V1.3. GPIO numbers identify RP2350A signals; framework aliases depend on the selected board definition.

4.1 General Pinout

Board label	RP2350A	Primary board role	Shared resource
TX0 / D1	GPIO18	UART-capable digital I/O	HSTX Data0 pair
RX0 / D0	GPIO19	UART-capable digital I/O	HSTX Data0 pair
D2	GPIO17	Digital I/O	HSTX Data1 pair
D3	GPIO16	Digital I/O	HSTX Data1 pair
D4	GPIO15	Digital I/O	HSTX clock pair
D5	GPIO14	Digital I/O	HSTX clock pair
D6	GPIO13	Digital I/O	HSTX Data2 pair
D7	GPIO12	Digital I/O	HSTX Data2 pair
D8	USB_DM (pin 51)	USB D- (native RP2350A USB PHY)	Same differential pair as the USB-C connector, via 22 Ω series resistor R10
D9	USB_DP (pin 52)	USB D+ (native RP2350A USB PHY)	Same differential pair as the USB-C connector, via 22 Ω series resistor R11
D10 / SS	GPIO21	Digital I/O	General-purpose
D11 / MOSI	GPIO23	Digital I/O	General-purpose
D12 / MISO	GPIO20	Digital I/O	General-purpose
D13 / SCK / LED	GPIO22	Digital I/O and user LED	BUILTIN1
A1 / D15	GPIO29 / ADC3	Analog-capable I/O	Also routed to HSTX connector
A2 / D16	GPIO27 / ADC1	Analog-capable I/O	Edge pad
A3 / D17	GPIO26 / ADC0	Analog-capable I/O	Edge pad
SDA / D18	GPIO8	I2C data	BMI270 bus
SCL / D19	GPIO9	I2C clock	BMI270 bus
D21	GPIO10	PDM clock	ICS-41350 clock
RGB	GPIO1	RGB-chain output	Three onboard WS2812 LEDs

Power and control positions include 3V3, 3EN, VBAT, VUSB, VIN, RST, and multiple GND pads. Their physical presence does not establish unreleased input or load limits.

4.2 Arduino NANO Pinout Compatibility

The board uses two parallel 15-position edge rows inspired by the Arduino Nano layout. This provides a familiar mechanical and labeling pattern, but it is not a statement of full electrical or shield compatibility.

Important differences include:

- RP2350A uses a 3.3 V logic domain.
- Several conventional positions have PULSAR-specific functions such as VBAT, 3EN, RGB, and microphone clock.
- GPIO12–GPIO19 are shared with the HSTX video route.
- D8 and D9 are not general-purpose GPIO: they carry the RP2350A's native USB D-/D+ signals (the same differential pair used by the USB-C connector), each through a 22 Ω series resistor. Do not drive them as digital I/O.
- Analog channel labels do not form a simple sequential GPIO order.
- The board includes bottom-side connectors and components that may conflict mechanically with some carriers.

Before installing the board into a Nano-style carrier, compare every power, reset, analog, and digital position and verify bottom-side clearance.

4.3 QWIIC Connector

J1 is a four-position, 1 mm-pitch right-angle connector carrying:

Signal	RP2350A connection	Function
GND	Ground	Common return
3.3 V	Regulated rail	Peripheral supply; available current not specified
SDA	GPIO24	External I2C data
SCL	GPIO25	External I2C clock

The QWIIC bus is separate from the GPIO8/GPIO9 bus used by BMI270. A controlled contact-number and mating-cable drawing is not included, so verify the physical orientation against controlled connector data when producing a harness.

4.4 MicroSD Connector

Socket signal	GPIO	Description
CLK	2	SDIO clock / SPI SCK
CMD	3	SDIO command / SPI MOSI
DAT0	4	SDIO data 0 / SPI MISO
DAT1	5	SDIO data 1
DAT2	6	SDIO data 2
DAT3	7	SDIO data 3 / SPI chip select

Socket signal	GPIO	Description
Detect	Dedicated socket contact	Card-detect handling depends on firmware routing
VDD	3.3 V	Card supply
GND / shields	Ground	Return and mechanical shield

Insert and remove a card only when filesystem activity has stopped. Software must flush and close files before power removal.

4.5 Battery Connections

JP1 is a two-position PH2.0 battery connector associated with VBAT and the battery return. Polarity markings are present on the bottom side. The MCP73831 charge controller is designed for a single-cell Li-Ion/Li-Polymer system, but the compatible cell, connector, polarity convention, charge current, and operating range are not specified by the available module-level documentation.

Do not connect a battery based solely on connector fit. Reversed polarity or an unsupported chemistry can damage the board or cell.

4.6 HSTX 22-pin Connector

J5 is a 22-position, 0.5 mm-pitch FFC/FPC connector. The schematic exposes D0–D7 (GPIO12–GPIO19), A0/GPIO28, A1/GPIO29, GPIO24/SDA, GPIO25/SCL, 3.3 V, and interleaved returns. HSTX video uses the eight GPIO12–GPIO19 signals as four TMDS pairs.

The flex-cable contact side, complete numbered contact table, cable length, and display adapter are not specified by a controlled mechanical drawing. Do not infer pin 1 from an unannotated photograph.

4.7 USB-C, BOOT, and Reset

The USB-C connector carries VBUS and the RP2350A USB data pair. The schematic shows the required USB-C configuration resistors and ESD protection. USB is the preferred programming and first-power interface.

BOOT selects the ROM USB boot path used for firmware recovery or initial programming. Reset restarts RP2350A while power remains applied. Exact button timing depends on the selected toolchain, but the common sequence is to hold BOOT while resetting or connecting USB, then release BOOT after enumeration.

4.8 SWD and Test Pads

Bottom-side pads expose SWDIO, SWCLK, 3.3 V reference, and GND for an external debug probe. SWD supports firmware loading and source-level debug when the toolchain and processor architecture are configured consistently.

Silkscreen erratum: The SWD VCC and GND labels are reversed. See Section 9.6 for the corrected mapping before connecting a debug probe.

Use short connections, share ground, and let the probe sense the target's 3.3 V rail. Do not use the reference pad to power the complete board unless the debug probe is explicitly rated and the board power path permits it.

5 Board Operation

The procedures below define the board workflow described by the technical wiki and the compilable sketches under `software/cpp_examples/`.

5.1 Getting Started with Arduino IDE

1. Install Arduino IDE 2.x.
2. Add the UNIT package index to **Additional Boards Manager URLs**: `https://raw.githubusercontent.com/UNIT-Electronics/Uelectronics-RP2040-Arduino-Package/main/package_Uelectronics_rp2040_index.json`
3. Install the UNIT Electronics RP2040/RP2350 package.
4. Select **UNIT Pulsar RP2350A** or an installed PULSAR-specific board definition.
5. Select the RP2350A chip variant, PSRAM CS GPIO0, and 8 MiB PSRAM.
6. Connect a USB-C data cable and select the enumerated serial/boot port.
7. Open the basic Blink sketch and upload it.

The repository examples were compile-checked with the UNIT RP2350 core using a 16 MiB flash and 8 MiB PSRAM configuration. Compile success confirms API and library compatibility; electrical operation also depends on the connected hardware and selected configuration.

5.2 First Power-up and Inspection

Use USB-C for initial power. Inspect component orientation, connector seating, solder bridges, and the microphone port before energizing the board. If available, use a current-limited source and monitor the 3.3 V rail.

Do not attach a battery, external VIN, HSTX adapter, microSD card, or QWIIC peripheral for the first power check. Confirm the power indicator and absence of unexpected heating, then establish USB enumeration and upload behavior.

5.3 BOOT and Reset Workflow

For a normal update, the selected framework may reset the board into its upload mode automatically. If automatic upload is unavailable:

1. Disconnect external driven signals.
2. Hold BOOT.
3. Press and release reset, or connect USB while BOOT is held.
4. Release BOOT after the ROM boot interface appears.
5. Copy or upload the generated firmware using the toolchain instructions.
6. Reset again to run the application from external flash.

SWD provides an alternative when debugging startup code or recovering a board whose USB application does not enumerate.

5.4 Recommended Bring-up Sequence

Order	Subsystem	What the step establishes
1	User LED	Core execution, GPIO20, and upload path
2	USB serial	Host communication and diagnostic output
3	ADC	Basic analog input path
4	RGB chain	GPIO1 timing and three-pixel order
5	BMI270 I2C	GPIO8/GPIO9 and onboard sensor communication
6	QWIIC I2C	GPIO24/GPIO25 and external expansion
7	PSRAM	GPIO0 selection, capacity detection, and allocation
8	microSD	SPI-compatible subset, card access, and filesystem
9	PDM	GPIO10/GPIO11 clock and audio capture
10	HSTX	GPIO12–GPIO19 pairing and display output
11	Combined application	Memory ownership and subsystem coexistence

Testing one subsystem at a time makes pin conflicts and library configuration errors easier to isolate.

5.5 C++ Example Collection

The `software/cpp_examples/` directory contains 28 complete sketches:

Group	Sketches	Coverage
Blink	2	basic and serial-debug LED operation
ADC	2	continuous reading and threshold indication
WS2812	3	color sequence, brightness, running pixel
I2C	7	scan, BMI270, EEPROM byte/block/dump/structure/multiple devices
microSD	4	initialize, read/write, list, log
HSTX	3	test pattern, drawing primitives, system monitor
Complete application	1	BMI270-controlled HSTX cube
PSRAM	5	detect, static/dynamic allocate, test, monitor

Group	Sketches	Coverage
PDM	1	capture PCM samples through the PDM library

The examples reconcile known wiki/API differences: user LED GPIO20, SDFS capacity through FSInfo, directory open mode, `flush()` return type, and release of `malloc()` memory through `free()`.

5.6 Memory Planning

Keep stacks, interrupt state, frequently accessed variables, and small DMA descriptors in internal SRAM. Place large framebuffers, logging queues, and historical data in PSRAM. Check every allocation and track free internal and external heap separately.

HSTX RGB565 graphics can consume a large buffer; microSD and USB also require working memory. Avoid allocating every maximum-size buffer at startup without a documented memory budget.

5.7 Storage Operation

Configure the microSD SPI subset on GPIO2, GPIO3, GPIO4, and GPIO7 for the current SDFS examples. Initialize the card, verify filesystem information, open only the files needed, call `flush()` after important writes, and close handles before removal or reset.

For data logging, buffer short records in RAM and write them in bounded batches rather than performing filesystem operations from an interrupt handler.

5.8 Video, Sensor, and Audio Coexistence

HSTX reserves GPIO12–GPIO19 and may generate continuous DMA/interrupt load. BMI270 uses GPIO8/GPIO9, PDM uses GPIO10/GPIO11, and microSD uses GPIO2–GPIO7, so the fixed signal sets do not overlap. They still compete for processor time, DMA channels, SRAM, PSRAM bandwidth, and power.

Use bounded update rates, partial display redraws, buffered audio acquisition, and nonblocking storage tasks. A combined application should expose health information over USB serial or RGB status rather than failing silently.

5.9 Troubleshooting Guide

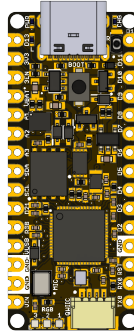
Symptom	Checks
No USB device	Data-capable cable, BOOT/reset sequence, host port, 3.3 V rail
Blink does not run	GPIO20 mapping, selected board/FQBN, successful upload
PSRAM unavailable	PSRAM CS GPIO0, 8 MiB option, allocation result
BMI270 not found	GPIO8/GPIO9 configuration, address selection, library
microSD fails	FAT32, GPIO2/3/4/7, SDFS configuration, card seating
PDM returns no data	GPIO10 clock, GPIO11 data, PDM buffer/callback setup
HSTX display blank	pair order, adapter/cable orientation, supported timing
Unstable combined app	internal heap, PSRAM use, blocking loops, pin conflicts

Record the exact board revision, package/core version, library versions, compiler output, power source, and reproduction steps for every bring-up issue.

6 Mechanical Information

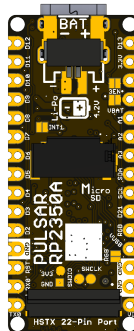
Hardware revision V1.3.0 uses a narrow development-board layout with two parallel edge-pad rows, four corner mounting holes, USB-C at one end, and the 22-pin HSTX connector at the opposite end.

6.1 Component-side Envelope



USB-C, pushbuttons, QWIIC, and the tallest top-side components require vertical and tool-access clearance. Edge castellations/pads must remain accessible for headers, carrier sockets, or direct soldering.

6.2 Bottom-side Envelope



The battery connector, microSD socket, microphone, and HSTX connector require bottom-side clearance. An enclosure must not obstruct the microphone acoustic port or prevent card and flex-cable insertion.

6.3 Mechanical Data Not Specified

A controlled board-outline drawing, PCB thickness, mounting-hole coordinates, keep-outs, connector insertion envelopes, and component height table are not present in the supplied package. Do not derive production dimensions from the rendered images.

An enclosure or carrier design requires independently controlled values for:

- overall board length, width, and thickness;
- edge-row pitch and row spacing;
- mounting-hole diameter and coordinates;
- USB-C, QWIIC, battery, microSD, and HSTX access envelopes;
- top/bottom maximum component height;

- microphone acoustic keep-out; and
- insertion/removal paths for card, battery plug, and FFC/FPC cable.

7 Company Information

Item	Information
Company	UNIT Electronics
Website	UNIT Electronics
Address	Salvador 19, Cuauhtémoc, 06000 Mexico City, CDMX, Mexico

The repository and Product Wiki in the next chapter provide the product's hardware, software, and reference documentation.

8 Reference Documentation

Reference	Location
Product repository	UNIT-Electronics-MX/unit_pulsar_rp2350a
Hardware reference	hardware/README.md
Technical wiki	PULSAR RP2350 wiki
Getting Started	Setup guide
C++ examples	software/cpp_examples
Board schematic	V1.3 schematic
RP2350 component datasheet	Raspberry Pi RP2350 Datasheet

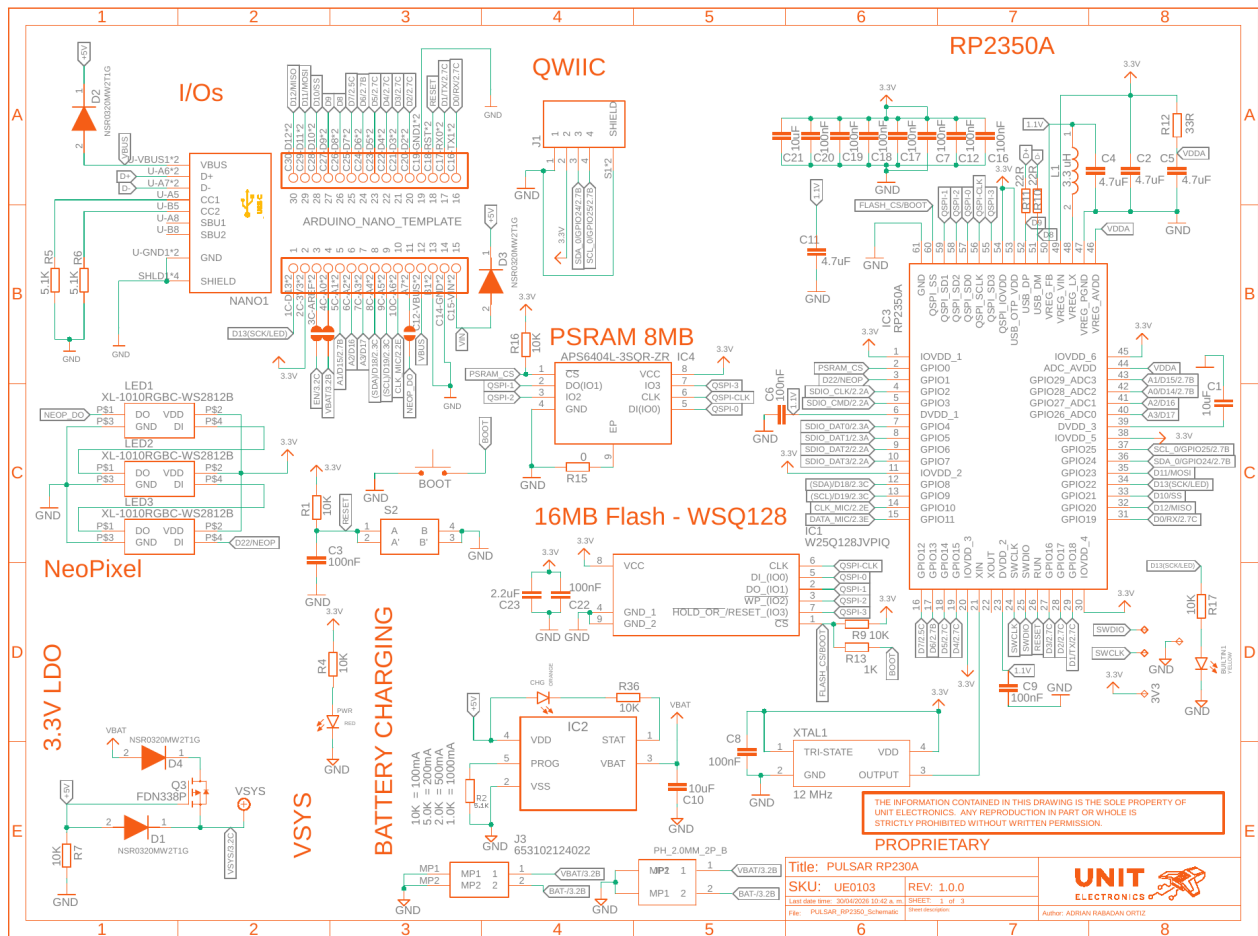
The wiki is the implementation guide for the board's intended operation, but it is not a source of module-level electrical ratings. The supplied legacy Product Reference DOCX is also excluded as a technical source because its content describes a PULSAR ESP32-H2 rather than this RP2350A board.

9 Appendix

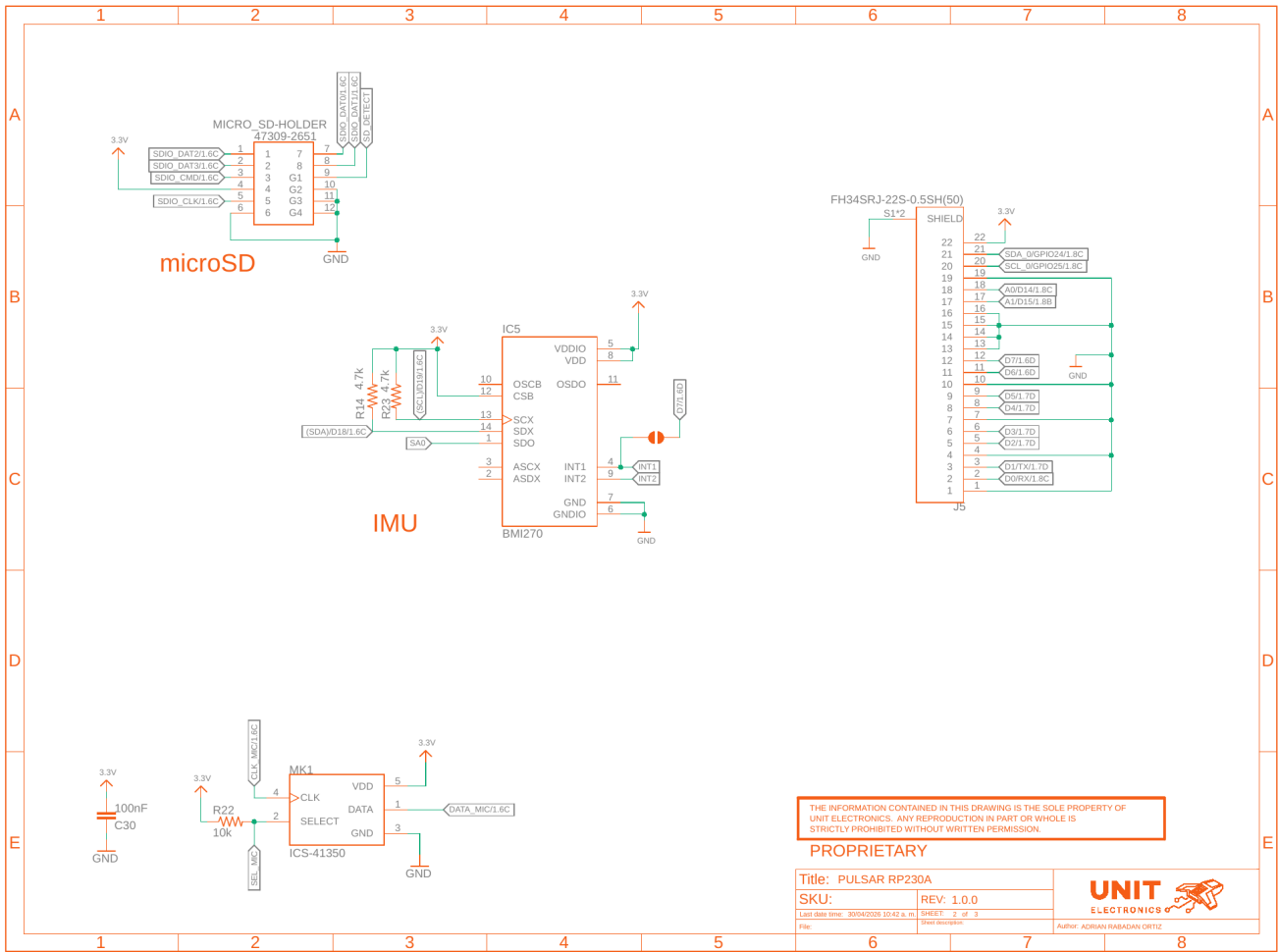
9.1 Schematic

The following three sheets are rendered from `unit_sch_v_1_3_0_pulsar_rp2350a.pdf`. The PDF remains the authoritative electrical drawing. The title block still contains PULSAR RP230A and revision 1.0.0; this label differs from the UNIT PULSAR RP2350 name and RP2350A controller described in this reference.

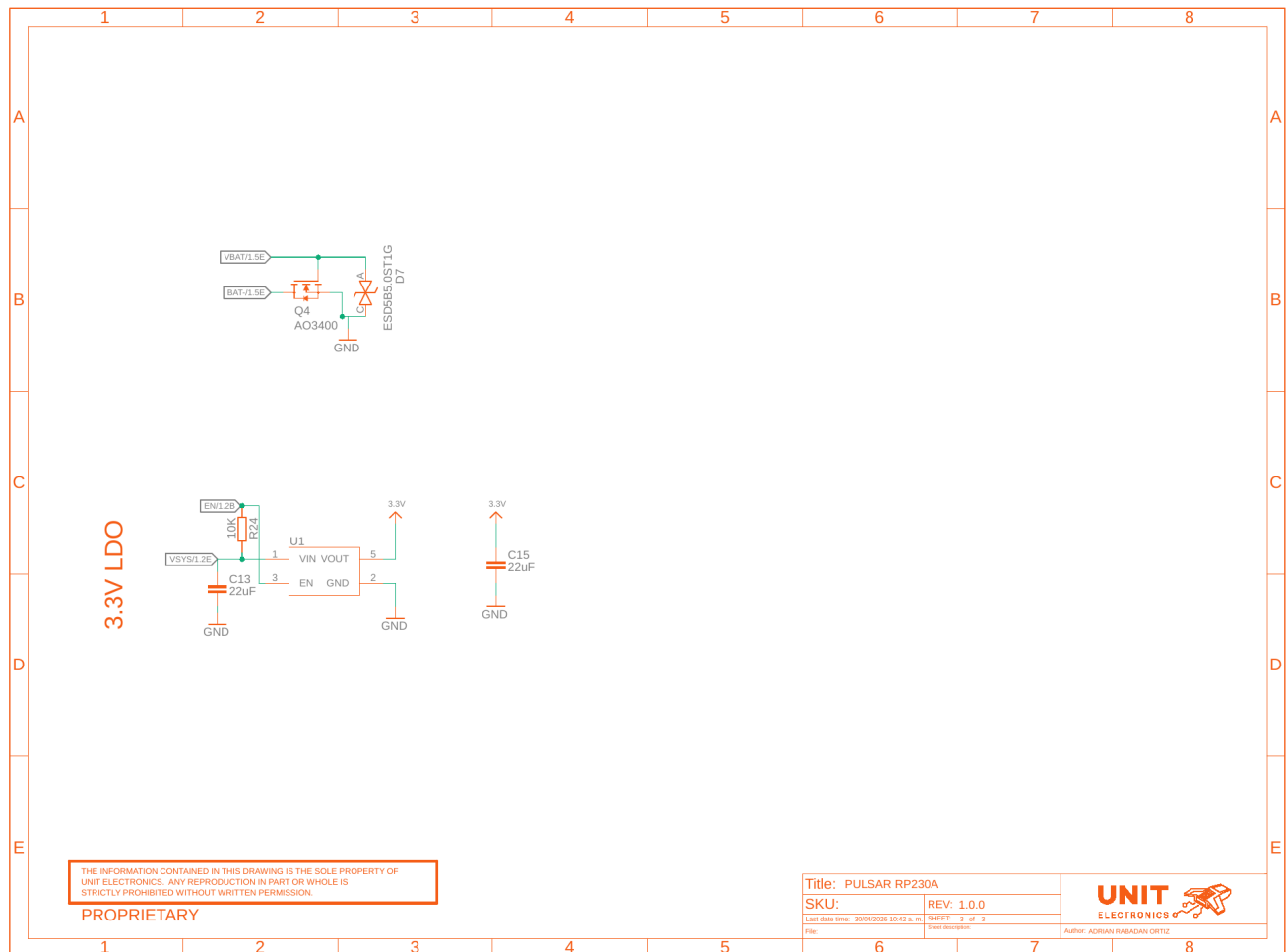
9.1.1 Processor, Memory, USB, I/O, and Power Entry



9.1.2 microSD, BMI270, HSTX Connector, and PDM Microphone



9.1.3 3.3 V Regulator



9.2 Technical References

1. The V1.3 schematic defines board connectivity and component designators.
2. The official Raspberry Pi RP2350 datasheet defines RP2350A component capabilities and limits.
3. The technical wiki defines intended subsystem operation and firmware workflow.
4. The compile-checked C++ example collection reconciles wiki code with the installed UNIT RP2350 core APIs.

9.3 Values Not Specified by the Technical Documentation

- A schematic title block matching UNIT PULSAR RP2350 V1.3.0
- Module VIN, VBAT, USB, and 3.3 V rail electrical limits
- Power-source priority, charge current, and compatible battery connection
- Complete numbered HSTX and QWIIC connector drawings
- D8 and D9 edge-pad disposition
- Board dimensions, mounting coordinates, and mechanical keep-outs
- Current consumption, thermal behavior, and validated interface rates
- Module-level runtime performance for the wiki and C++ examples

- Complete pinout, topology, and dimension drawings

9.4 Document Control

Field	Value
Product	UNIT PULSAR RP2350
Product family	UNIT DevLab ecosystem
Hardware revision	V1.3.0
Product Reference	Version 0.1.0
Publication date	2026-08-03
Firmware	28 compile-checked examples

9.5 Source Inconsistencies

- The board uses an RP2350A, but the schematic title block says PULSAR RP230A and revision 1.0.0.
- The supplied legacy Product Reference DOCX describes an ESP32-H2 board and is not applicable to UNIT PULSAR RP2350.
- Wiki microSD examples use SPI mode, while the V1.3 schematic exposes a full four-bit SDIO group. Both are represented correctly: SPI uses the compatible CLK/CMD/DAT0/DAT3 subset.
- The wiki's original Blink definition used GPIO22, while the V1.3 schematic connects D13 / BUILTIN1 to GPIO20. Repository examples follow GPIO20.

9.6 Hardware Errata

SWD connector: reversed VCC and GND silkscreen labels. The VCC and GND labels printed beside the SWD pads on the UNIT PULSAR RP2350 board are swapped. Use the following corrected mapping when connecting a debug probe:

Printed silkscreen label	Actual electrical connection
VCC	GND (ground)
GND	VCC (3.3 V reference)

Verify the pad connections against the schematic before applying power. Following the incorrect silkscreen labels reverses VCC and GND and may damage the board or the debug probe. Follow the SWD connection guidance in Section 4.8.